MILITARY SPECIFICATION

RADAR SET AN/SPS-( )

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for construction and testing of a combined type II and type IV surface search and navigational radar for use aboard navy ships and craft, which will be designated AN/SPS-( ). IFF capability will not be required. Type II and IV radars are defined for purposes of this specification as surface search radars suitable for small boat and auxiliary operation.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-17 - Cables, Radio Frequency, Coaxial, Dual Coaxial, Twin Conductor and Twin Lead.
MIL-T-27 - Transformers and Inductors (Audio, Power, and High-Power Pulse), General Specification for.
MIL-E-75 - Electron Tubes, Preparation for Delivery of.
MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-F-3922/53 - Flanges, Waveguide (Cover) (Square, 4 Holes) (With Seal Groove).
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-C-15305 - Coils, Fixed and Variable, Radiofrequency; General Specification for.
MIL-S-15743 - Switches, Enclosed, Rotary, Snap Action.
MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore; General Specification for.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of.
MIL-R-22732 - Reliability Requirements for Shipboard Electronic Equipment.
MIL-I-22960 - Indicators, Radar, Naval Ship and Shore, General Specification.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment. (Type I - Environmental and Type II - Internally Excited).
MIL-STD-242 - Electronic Equipment Parts (Selected Standards).
MIL-STD-415 - Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-471 - Maintainability Demonstration.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781 - Reliability Tests Exponential Distrubution.
MIL-STD-882 - System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for.
MIL-STD-1399 - Interface Standard for Shipboard Systems, Section 103 - Electric Power, Alternating Current.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:18Z
Check the source to verify that this is the current version before use.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.
MIL-HDBK-239 - Navy Standard Hardware Program Application Handbook.
MIL-HDBK-472 - Maintainability Prediction.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

NAVSHIPS

0967-008-9000 - Test Equipment Index to NAVSHIPS Prime Electronics Equipments.
0967-052-8280 - Engineered Restoration Procedure for AS-1091/SPS-43 and AS-943/SPS-29 Radar Antennas.
0967-309-3010 - Design of Shock and Vibration Resistant Electronic Equipment for Shipboard use.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting officer.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

3.2 General. The AN/SPS-( ) shall be in accordance with the requirements of MIL-E-16400 where specified herein. Microminiaturization will be allowed as part of this specification. Functional modular circuit construction shall be incorporated in the design of the receiver-transmitter and the control indicator units of the AN/SPS-( ). The Government encourages use of Navy Standard Hardware Program (SHP) modules wherever practicable and are defined in MIL-HDBK-239. No part shall be stressed more than 50 percent of rated value at nominal operation, with the exception of tubes which shall not be stressed more than 80 percent.

3.2.1 Controls. Controls shall conform to MIL-E-16400. Tactical identification of controls shall be as specified in MIL-STD-242. Toggle switches shall be mounted vertically with the "ON" position upward.

3.2.2 Operating modes. The system switching shall furnish power only to those units requiring power for the particular mode of operation. Units not requiring power for a given operating mode shall have no power applied in that mode. This applies to parts, subassemblies, assemblies and units. The modes of system operation shall be as follows:

(a) OFF
(b) STANDBY
(c) RADIATE

3.2.3 Installation kit. Any special waveguide parts or any other items that are special or unique for installation shall be supplied by the contractor with each radar in a separate installation kit.

3.2.4 Special tools and test equipment. No special tools or test equipment shall be required for the repair, maintenance, or alinement of Radar Set AN/SPS-( ). Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)). Selection of portable test equipment shall be from Publication 0967-008-9000.

3.2.5 Illumination. Illumination of controls and indicators shall be in accordance with MIL-E-16400, except as otherwise specified herein. Illumination in spaces where dark adaptation must be maintained shall be in accordance with the requirements for dark adaption of MIL-I-22960.

3.2.6 Electromagnetic interference control. Radar Set AN/SPS-( ) shall conform to the electromagnetic interference control requirements of MIL-E-16400. The applicable subclass shall be class IA or IB of MIL-STD-461.

Source: https://assist.dla.mil -- Downloaded: 20260-19T22:18Z
Check the source to verify that this is the current version before use.

3.3 Equipment composition. Radar Set AN/SPS-( ) shall consist of the major units shown in table I. The only tubes permitted for use are the transmitter magnetron, hard tube modulator, cathode ray tube (CRT), receiver TR tube, and local oscillator, (see 3.4.2.1.4 and 3.4.2.2.7).

Table I - Equipment composition.

Unit No.	Description	Paragraph
1	Antenna Unit	3.4.1
2	Receiver-Transmitter (R/T)	3.4.2
3	Control Indicator (CI)	3.4.3
4	Power Supply (P/S)	3.4.4
5	Antenna Safety Switch	3.4.5

3.3.1 Volume and weight. The maximum volumes and weights shall be as shown in table II.

Table II - Maximum volume and weights.

Unit	Volume Cubic Feet	Weight Pounds
Antenna:		
Type I	-	1/ 65
Type II	-	1/ 95
Receiver-Transmitter (R/T)	1.2	45
Control Indicator (CI)	4.8	75
Power Supply (P/S)	0.6	30
Antenna Safety Switch	.2	5

1/ The maximum antenna weights include the antenna and pedestal.

3.4 Performance.

3.4.1 Antenna unit. The antenna unit (antenna plus antenna drive mechanisms) shall be capable of directly interfacing with the receiver transmitter and control indicator specified in 3.4.2 and 3.4.3, respectively. The antenna shall be one of the following types, as specified (see 6.1.1):

Type I - Antenna enclosed in a radome
Type II - Antenna with no radome

3.4.1.1 Type I. The type I antenna unit shall be an antenna 3 feet maximum in length enclosed in a 42 inch maximum outside diameter radome. The radome shall be equipped with non-metallic handles mounted 180 degrees apart to facilitate handling. A security cable or chain attached to the radome with captive screws or fittings and fastened to the housing or other convenient support shall be provided to prevent dropping the radome.

3.4.1.2 Type II. The type II antenna unit shall be an antenna with a length greater than 6 feet but less than 9 feet, with no radome.

3.4.1.3 General requirements.

3.4.1.3.1 Antenna arrays. Each antenna option shall be a horizontally slotted array.

3.4.1.3.2 Wind speed. The antenna unit shall operate normally in winds having a relative velocity of 80 knots with an equivalent load specified in 3.4.1.3.3. The antenna systems shall not be damaged in winds with relative velocities up to 130 knots, whether drive motors in the pedestals are energized or not.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T22:18Z
Check the source to verify that this is the current version before use.

3.4.1.3.3 Icing. The antenna unit shall conform to the requirements of MIL-E-16400 for icing and, in addition, shall not be damaged by an ice loading of 7 pounds per square foot (lb/ft^2).

3.4.1.3.4 Swing circle (type II). The swing circle of the type II antenna shall be 114 inches maximum.

3.4.1.3.5 Peak power. All parts of the antenna unit concerned with the transmission and radiation of radar RF energy shall be capable of operating with peak power of 30 kilowatts (kW) minimum without arcing. Operation shall be compatible with transmitted peak power.

3.4.1.3.6 Voltage standing wave ratio (VSWR). The overall VSWR of the antenna unit as measured at the base of the pedestal shall be not greater than 1.3 to 1 over the frequency band specified in 3.4.1.3.7 at all azimuth positions of the rotary joint.

3.4.1.3.7 Frequency. The antenna unit shall operate within the frequency band of 9000 ± 1000 megahertz (MHz).

3.4.1.3.8 Horizontal beamwidth. The width of the main beam at the half-power points for the type I antenna shall be 2.6 degrees maximum and 0.9 degree maximum for the type II antenna 9 feet in length, at the frequency specified in 3.4.1.3.7. Type II antennas from 6 to 9 feet may have horizontal beamwidths proportionally larger than 0.9 degree. All sidelobes inside of 10 degrees either side of the center of the main beam shall be down at least 23 decibel (dB) from the peak. Outside 10 degrees, they shall be down a minimum of 28 dB.

3.4.1.3.9 Vertical beamwidth. The maximum energy shall be obtained midway between the half-power points within 0.5 degree. The point of maximum radiation shall be directed at zero degrees elevation. The half-power beamwidth shall be not less than 20 degrees, over the frequency specified in 3.4.1.3.7.

3.4.1.3.10 Bearing accuracy. The bearing accuracy of the electromechanical positioning system shall be within 0.25 degree relative.

3.4.1.3.11 Mounting. The antenna unit shall be provided with a mounting flange to fit a standard 16-1/2 inch diameter bolt circle or a standard 2-1/2 inch stand-pipe.

3.4.1.3.12 Accessibility. Accessibility shall be in accordance with MIL-E-16400. Access covers shall be secured by a length of chain or, if the weight is over 3 pounds, shall be hinged with means provided for securing in the open positions. Covers shall be restrained with captive screws or fittings.

3.4.1.3.13 R F Coupler. A flange cover (UG-39/U) Part No. M3922/53-001 of MIL-F-3922/53 shall be provided at the base of the antenna unit. The flange cover shall be recessed and shall have a window.

3.4.1.3.14 Ships heading marker (SHM). An SHM circuit shall provide a displayed signal when the antenna is trained through zero degree relative bearing. This fixed reference, available at all times, shall require no synchronization circuitry or adjustments after installation.

3.4.1.3.15 Rotation. The antenna drive mechanism shall be capable of rotating the antenna in azimuth at 20 revolutions per minute (r/min), or better, under the environmental conditions specified herein, at nominal line voltage and frequency.

3.4.1.3.16 Noise. During rotation and operation, the audible noise emanating from the antenna shall be not greater than the acceptance levels of airborne grade D of MIL-STD-740. During normal operation, the noise levels produced by the radar set shall be not greater than the acceptance levels of airborne grade C of MIL-STD-740.

3.4.2 Receiver transmitter unit. The receiver transmitter unit shall be capable of either bulkhead or deck mounting and a base plate shall be provided for the unit. The R/T unit shall be a separate unit, not contained in the antenna unit.

3.4.2.1 Receiver.

3.4.2.1.1 Receiver composition. The receiver shall be part of the R/T unit.

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T22:18Z
Check the source to verify that this is the current version before use.

3.4.2.1.2 Receiver characteristics. The total receiver circuit through the IF amplifier shall have the following characteristics:

(a) Noise figure - 11 dB, or less.
(b) Bandwidth - Compatible with short and long pulsewidths.

3.4.2.1.2.1 IF amplifier. The IF amplifier shall be a linear or switchable linear/logarithmic amplifier.

3.4.2.1.3 Mixers. Signal and local oscillator mixers capable of operating over the required range of frequencies shall be provided. The mixer assemblies shall have a noise figure compatible with overall receiver noise figure of 11 dB, or less. The crystals shall be located for ease of replacement. Protection circuitry shall be provided for the mixers.

3.4.2.1.4 Local oscillator tuning. A tuning circuit shall maintain the local oscillator at the predetermined operating frequency above the transmitter frequency.

3.4.2.1.5 Sensitivity time control and main bang suppression (STC and MBS). The transmitter main bang shall be blanked out in both the long and short pulse modes of operation. The STC shall be effective to 12 nautical miles and shall be capable of reducing the receiver gain by at least 70 dB. The STC shall provide approximately 12 dB per octive attenuation to the receiver response.

3.4.2.1.6 Fast time constant (FTC) function. The FTC function ON-OFF switch shall be located on the CI unit.

3.4.2.1.7 Trigger generator. The trigger generator shall provide not less than one internal and one external trigger.

3.4.2.1.8 Fuses and test points. Fast acting fuses and a test point for each voltage shall be installed (see 3.4.11.6). Circuit test points shall be provided in accordance with MIL-STD-415.

3.4.2.1.9 Frequency. The receiver shall operate within the frequency band of 9000 ± 1000 MHz.

3.4.2.1.10 Frequency tuning. The receiver shall have a manual frequency control.

3.4.2.1.11 Range.

3.4.2.1.11.1 Minimum range. The minimum range of the radar shall be 25 yards, or less.

3.4.2.1.11.2 Range resolution. The range resolution shall be 20 yards, or less.

3.4.2.1.11.3 Range accuracy. The range accuracy shall be plus or minus 1 percent of the range scale in use.

3.4.2.2 Transmitter. The transmitter unit shall contain the following parts:

(a) Duplexer (see 3.4.2.2.5).
(b) Magnetron (see 3.4.2.2.6).
(c) Modulator (see 3.4.2.2.7).
(d) Power supplies as required.
(e) Time totalizing meter (see 3.4.11.1).

3.4.2.2.1 Frequency. The frequency shall be 9000 ± 1000 MHz, and shall be determined by the frequency of the magnetron selected.

3.4.2.2.2 Pulse durations. The RF output pulses, when measured at the 3 dB level from the peak shall be as follows:

Short pulse: 0.05 to 0.10 microsecond (us)
Long pulse: 0.5 to 1.0 us

3.4.2.2.3 Repetition rates. The long pulse repetition rate shall be 800 ± 100 pps. The short pulse repetition rate shall be 2550 ± 250 pps.

Source: https://assist.dla.mil -- Downloaded: 20260-19T22:18Z
Check the source to verify that this is the current version before use.

3.4.2.2.4 Filament voltages and filament switching. Time delays shall be provided for the protection of filaments, magnetron, or other parts during OFF-STANDBY-RADIATE switching. OFF to RADIATE time shall be not greater than 3 minutes. STANDBY to RADIATE shall be instantaneous.

3.4.2.2.5 Duplexer. The duplexer portion of the transmitter shall be capable of continuous operation over the transmitter frequency band at the maximum power level as specified herein. One way duplexer losses shall not exceed 0.5 dB.

3.4.2.2.5.1 The duplexer shall be capable of protecting the receiver in the event of open or short circuit in the transmission waveguide.

3.4.2.2.5.2 The duplexer protection circuit shall be consistent with the peak power output of the transmitter tube.

3.4.2.2.6 Magnetron. The transmitter shall use a magnetron type tube selected in accordance with the parts requirement of MIL-E-16400.

3.4.2.2.6.1 Peak power output. The peak power output of the magnetron shall be not less than 10 kw at any frequency in the band and at any steady state voltage within the stipulated voltages.

3.4.2.2.7 Modulator. A solid state or hard tube type modulator shall be provided to drive the magnetron and shall be compatible with the transmitter power output, pulse duration, and repetition frequencies specified herein. The modulator shall be protected by a bypass diode and despiking circuit. Test points conforming to MIL-STD-415 shall be included for viewing high voltage pulses at a safe level.

3.4.3 Control indicator.

3.4.3.1 Display. The display shall consist of a CRT plan position indicator (PPI) display, 9 inches minimum in diameter and shall conform to MIL-I-22960, except as otherwise specified herein. The CI unit shall contain the controls and indicators needed to control and monitor the performance of the AN/SPS-( ) radar.

3.4.3.1.1 Range scales. The following range scales shall be provided:

1/4, 3/4, 1-1/2, 4, 12, 24, and 36 (nautical miles)

3.4.3.1.2 Fixed range markes. Marker intervals shall be as shown in table III. The accuracy of the marks shall be within plus or minus 1.0 percent. The intensity of the marks shall be adjustable.

Table III - Range marker intervals.

Range scale (nautical miles)	Marker interval (nautical miles)
1/4	1/8
3/4	1/4
1-1/2	1/4
4	1
12	2
24	4
36	6

3.4.3.2 Circuit construction. The CI shall use functional modular plug-in printed circuit boards. The circuit boards shall contain, as a minimum, the following specific functions:

(a) Range scales
(b) Range marks
(c) Video
(d) PRF generator
(e) Time delay
(f) Sweep drive
(g) True bearing
(h) Variable range mark

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:18Z
Check the source to verify that this is the current version before use.

3.4.3.3 Variable range mark. The variable range mark shall be generated digitally, and shall be variable from 25 yards to 36 nautical miles. The position of the variable marker shall be indicated by a three digit counter. The range counter and indicator shall have 0.01 mile increments for range scales below 5 miles and 0.1 mile for range scales 5 miles and over. Accuracy of the variable range mark shall be plus or minus 1.0 percent and shall agree with the fixed range markers within plus or minus 1 marker pulse width. The marker shall be adjustable in intensity.

3.4.3.4 Sweep linearity. The PPI sweep deflection shall attain a slope of 50 percent of the desired value in 5 percent of the trace length or within 0.3 us, whichever is the greater.

3.4.3.5 True/Relative bearing. True or relative bearing data shall be provided and shall be selectable and displayed by the operation of a true/relative switch. Bearing resolution shall be 1 degree and bearing accuracy shall be plus or minus 1 degree.

3.4.3.6 Azimuth indication. The CRT deflection coil shall be of a fixed nonrotating design. There shall be no rotating parts. The CRT shall be enclosed in metal shields to eliminate possible sweep interference from external magnetic fields.

3.4.3.7 Controls, indicators, and panel layout. Controls, indicators, and panel layout shall be in accordance with MIL-E-16400. The control indicator shall contain but is not limited to the following typical controls and indicators:

(a) Ship's heading marker (ON-OFF)
(b) Radar status light (OFF-STANDBY-RADIATE)
(c) PPI intensity
(d) Video gain
(e) Receiver tune
(f) Sensitivity time control (STC)
(g) Fast time contant - FTC (ON-OFF)
(h) Range selector (Range and MARKS)
(i) Range marks (ON-OFF)
(j) Variable range mark (ON-OFF)
(k) Antenna rotation (ON-OFF)
(l) Panel light dimmer
(m) Vertical centering offset
(n) Horizontal centering offset
(o) Ready light
(p) PPI sweep intensity
(q) Fixed range marker intensity
(r) Variable range marker intensity
(s) Ship's heading marker intensity
(t) Bearing cursor
(u) "Man Aloft" light
(v) Control indicator switch (ON-OFF)

3.4.3.7.1 Panel layout. The front panel controls and indicators shall be accessible and visible.

3.4.3.7.2 Ready light. The READY light shall illuminate when a time delay of 3 minutes, or less, elapses after the AN/SPS-( ) is switched from OFF to STANDBY.

3.4.3.7.3 Illumination. The front panel shall be back lighted. Illumination shall conform to the requirements of MIL-E-16400 for dark adapted areas.

3.4.3.8 Viewing hood. A viewing hood shall be supplied to facilitate day time use of the display.

3.4.3.9 Non-glare filter. A non-glare, easily removable, orange-colored filter shall be installed over the display for "night vision" protection of the operator. Storage shall be provided for the filter when not in use.

3.4.3.10 CRT phosphor protection. A time delay shall be provided to prevent burning a hole in the phosphor of the CRT during turn-on or turn-off.

3.4.3.11 Mounting. The control indicator configuration shall permit table, bulk-head, and overhead mounting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:18Z
Check the source to verify that this is the current version before use.

3.4.3.11.1 Circuit access. Circuit access shall be readily available with respect to any of the three mounting installations.

3.4.3.12 Test points. Circuit test points for maintenance and upkeep shall be provided in accordance with MIL-STD-415.

3.4.4 Power supplies. The modulator high voltage power supply shall include circuits which disable the supply during pulsewidth switching and under the fault conditions specified herein. No power supply shall be damaged by a direct short of the output.

3.4.5 Antenna safety switch. A separate watertight radar antenna safety switch for opening the radiate interlock circuit and removing power to the antenna drive motor shall be provided as part of the equipment and shall have provisions for mounting near the access to the antenna platform. The two switch positions shall be plainly marked "OPERATE" and "DISABLE". The switch shall be in accordance with MIL-S-15743. The switch shall light the "MAN ALOFT" lamp in the Control Indicator when in the "DISABLE" position.

3.4.6 Maintainability. The equipment repair time (ERT) specified for all units shall be not greater than 1/2 hour. Maximum use of a module replacement concept at the organizational (shipboard) level shall be assured by a design emphasis on avoidance of piece part repair on board ship.

3.4.6.1 Maintainability prediction. A maintainability prediction shall be made and submitted in accordance with procedure II, part A of MIL-HDBK-472 (see 6.1.2). Mechanical items shall be included in the prediction. Assumptions made in performing the prediction shall be explicity stated.

3.4.7 Reliability. The reliability requirement for Radar Set AN/SPS-( ) shall be a specified mean-time-between-failure (MTBF) of 1000 hours minimum.

3.4.7.1 Reliability prediction. A reliability program, stress analysis, and MTBF prediction of Radar Set AN/SPS-( ) shall be performed in accordance with MIL-R-22732, except that failure rates for semiconductor devices (for example, diodes, transistors, SCR's, integrated circuits, and others) shall be obtained from MIL-HDBK-217 (see 6.1.2). Other failure rates may be selected from MIL-HDBK-217, FARADA (BUWEPS failure rate data) or other government approved sources. The contractor shall specify the source used for each failure rate selected and shall submit the source information, all MTBF calculations, and data sheets with the prediction. All failure rates shall be adjusted for a shipboard environment.

3.4.8 System safety. The contractor shall develop and maintain an effective system safety program that is planned and integrated into all phases of design, production, and testing of equipment (see 6.1.2). The system safety program shall provide a disciplined approach to indentify hazards and prescribe corrective actions in a timely cost effective manner. The system safety program tasks shall be specified in a formal plan (system safety program plan (SSPP)). The plan shall include requirements to be imposed on each subcontractor to assure compatibility with the system safety program for the equipment. MIL-STD-882 shall be used as guidance for preparing the SSPP.

3.4.8.1 Safety analyses. Safety analyses shall be performed to indentify hazardous conditions for the purpose of their elimination or control. Analysis shall be made to examine the equipment, subsystems, units, parts, and their interrelationship to include logistic support, training, maintenance and operational environments.

3.4.9 Environmental.

3.4.9.1 Temperature. Radar Set AN/SPS-( ) shall conform to the temperature requirements of the following classes of MIL-E-16400, as applicable.

(a) Range 2 - for equipment used above deck (exposed, unsheltered).
(b) Range 4 - for equipment used below deck (sheltered controlled environment).

3.4.9.2 Humidity. Radar Set AN/SPS-( ) shall conform to the humidity requirements of MIL-E-16400.

3.4.9.3 Heat removal. Heat removal shall be by air only in accordance with MIL-E-16400. Water cooling shall not be used.

Source: https://assist.dla.mil -- Downloaded: 202640-19T22:18Z
Check the source to verify that this is the current version before use.

3.4.9.4 Shock. Radar Set AN/SPS-( ), including antenna, shall meet the performance characteristics of this specification before and after being subjected to the following shock test requirements of MIL-S-901:

(a) Grade A of shockproofness.
(b) Equipment classification - Deck mounted, except for antenna.
(c) Class required - Class I or II, as applicable.
(d) Test classification - light- or medium-weight, as applicable.
(e) Type - "A", except "B" for antenna unit.
(f) There shall be no degree of permanent functional impairment.
(g) Test fixture as specified in MIL-S-901 or a substitute approved by the command or agency concerned.

3.4.9.5 Vibration. Radar Set AN/SPS-( ) shall perform normally under the vibration conditions as specified for type I of MIL-STD-167-1 from 4 through 50 Hz.

3.4.9.6 Inclination. Radar Set AN/SPS-( ) shall conform to the 45 degree inclination requirements of MIL-E-16400.

3.4.9.7 Enclosure. The above-deck equipment (antenna unit) shall conform to the watertight enclosure requirements of MIL-STD-108. Provisions shall be made for protecting the above-deck unit from damaging effect of shipboard stack gases. The below-deck units shall conform to the spraytight enclosure requirement of MIL-STD-108.

3.4.9.7.1 Structural integrity. Structural integrity shall be in accordance with MIL-E-16400. If lower resonant frequencies are encountered, these shall be eliminated by redesign and fabrication techniques using the guidelines of Publication 0967-309-3010.

3.4.10 Construction. The unit and subunit construction shall permit rapid assembly and disassembly. The equipment shall conform to the modular construction requirements of MIL-E-16400. Removal of a particular circuit card for repair/replacement shall not necessitate removal of other cards or individual parts not mounted on cards.

3.4.10.1 Material. The material used in the construction of Radar Set AN/SPS-( ) shall conform to MIL-E-16400, except as otherwise specified herein.

3.4.10.1.1 Aluminum or any other lightweight material approved by the command or agency concerned may be used where the weight of units can be significantly reduced without penalizing strength, machine accuracy, or resistance to wear and corrosion. Use of magnesium will not be permitted.

3.4.10.1.2 All external hardware shall be made of corrosion-resisting metals as specified in MIL-E-16400, and shall meet the salt fog (spray) requirements of MIL-E-16400.

3.4.10.2 Cables and transmission lines. Cables and transmission lines shall be as specified in MIL-E-16400, except as otherwise specified herein. All special interconnecting cables and connectors shall be furnished by the contractor. All special interconnecting cables shall be approved by the command or agency concerned prior to use. Coaxial cables shall be in accordance with MIL-C-17, and shall be selected from those listed in MIL-STD-242. To minimize cable weight, non-armored cable shall be used.

3.4.10.2.1 The following external interconnecting circuits shall be run in a cable or cables separate from all other circuits, and all terminal boards and cable entrances shall be designed and arranged to accommodate this division of circuits:

(a) Alternating current (a.c.)power circuits carrying over 0.5 ampere (A).
(b) Circuits connected to a commutated device.

3.4.10.3 Conductor identification. Conductor identification shall be in accordance with MIL-E-16400, except that colored wires, in addition to white wires, may be used for miscellaneous circuits to enhance circuit tracing.

3.4.10.4 Welding. Welding shall be in accordance with MIL-E-16400.

3.4.10.5 Corrosion protection/corrosion resisting treatment and painting. Corrosion protection/corrosion resisting treatment and painting shall be as specified in MIL-E-16400, except as otherwise specified herein.

3.4.10.5.1 The use of chemical films in accordance with MIL-C-5541 on aluminum parts will be permissible where anodizing, plating, or other metal finishes are not practical because of electrical properties or part size.

Source: https://assist.dla.mil -- Downloaded: 202640-19T22:18Z
Check the source to verify that this is the current version before use.

3.4.10.5.2 Below deck units. The finish coats for below-deck units shall be gray enamel conforming to type III, class 2 of MIL-E-15090. Hermetically sealed assemblies or other parts which cannot withstand the 121 degrees celsius (°C) (250°F) curing temperature required for type III finish shall be finished with type II, class 2 enamel of MIL-E-15090.

3.4.10.5.3 Above deck units. The above-deck units shall be finished with black epoxy paint to minimize the effects of corrosion from stack gases in accordance with Publication 0967-052-8280, except sand blasting is not required.

3.4.10.6 Workmanship. Workmanship shall be consistent and in accordance with MIL-E-16400.

3.4.10.7 Identification and marking. Identification and marking shall be in accordance with MIL-E-16400.

3.4.11 Parts and assemblies.

3.4.11.1 Time totalizing meters. Time totalizing meters shall be provided in accordance with MIL-E-16400, and shall independently record standby and radiate power conditions.

3.4.11.2 Transformers and inductors. Transformers and inductors shall conform to the requirements specified in MIL-E-16400, except that they shall conform to grade 4 or 5, class R, life expectancy X of MIL-T-27, as applicable.

3.4.11.3 Intermediate frequency and radio frequency (RF) transformers and coils. Intermediate frequency and RF transformers and choke coils shall conform to grade 1 or 2, class O of MIL-C-15305 and shall be selected from MIL-STD-242.

3.4.11.4 Pilot light indicators. Pilot light indicators shall be provided on each major unit to indicate the operation of all major sequences. A mechanical dimmer of the type which does not permit the control indicator pilot lights to be completely extinguished shall be provided for all pilot lights, except those used for bearing-dial or meter illumination. Press-fit bezels and bull's eyes shall not be used.

3.4.11.5 Indicator lamps. Indicator lamps shall be provided for circuit functions in accordance with MIL-E-16400, except for dark adapted areas (see 3.2.5).

3.4.11.6 Fuses and fuseholders. Fuses and fuseholders shall conform to the requirements specified in MIL-E-16400.

3.4.11.7 Connectors. Connectors shall be in accrodance with the requirements specified in MIL-E-16400, except as otherwise specified herein. The mating connectors shall be furnished with all connector receptacles or plugs. Connectors shall be used in lieu of removable blank plates and stuffing tubes for all below deck units.

3.4.11.7.1 Connectors for antenna control cables. Antenna control cables shall be fitted with small quick-disconnect plugs selected from MIL-STD-242. The plugs and mating connectors shall be waterproof.

3.4.11.8 Interchangeability. Interchangeability of parts shall conform to MIL-E-16400.

3.4.11.9 Proprietary assemblies. No proprietary modules, assemblies, or units shall be furnished as part of Radar Set AN/SPS-( ).

3.4.11.10 Printed wiring. Printed wiring shall be in accordance with the requirements specified in MIL-E-16400. On those printed wiring boards where space does not permit the reference designations to be printed on the board, the designations shall be located on the surface (module cover) adjacent to the board and the information shall be included in short form in the technical manual. Each printed wiring board shall be identified on or adjacent to the board as to the circuit function.

3.4.12 Power requirements.

3.4.12.1 Power supply. Power supply circuits shall conform to and operate from 115 V, 60 Hz, two-wire single phase type I power of section 103 of MIL-STD-1399 and from 12 volts direct current (d.c.) ± 10 percent, 24 volts d.c. ± 10 percent, or 36 volts d.c. ± 10 percent.

Source: https://assist.dla.mil -- Downloaded: 202640-19T22:18Z
Check the source to verify that this is the current version before use.

3.4.12.1.1 Plug-in modules incorporating the appropriate d.c. power supply circuits to conform to the craft/ship power lines shall be provided as part of the R/T or as a separate unit.

3.4.12.1.2 Power convertors shall be solid state.

3.4.12.1.3 The maximum power consumption shall be as follows:

(a) 300 watts (W) for d.c. power inputs.
(b) 350 W for a.c. power input.

3.4.12.2 Manually controlled switch. A manually-operated switch, shall be provided to disconnect all power including synchro signals, from Radar Set AN/SPS-( ).

3.4.12.3 Ship line overvoltage protection. Overload devices shall be provided which will protect the radar circuits from variations in the ship power.

3.4.12.4 Reverse voltage polarity protection. Reverse voltage polarity protection shall be provided in the radar d.c. power system.

3.4.12.5 Voltage regulator. A solid state voltage regulator shall be provided for voltages supplied by the ship d.c. power systems.

3.4.12.6 Control circuit. Electronically-operated power contactors shall be incorporated in the system as required to control the low voltage power supplies, system filaments, high voltage power supplies, and radar antenna. The contactors shall be interlocked via mechanical, temperature, air flow, or over-load interlocks contained in the various units of the systems. Fault interrupting devices shall be provided to protect the control and power circuits.

3.5 Repair parts. Repair parts shall be furnished as specified (see 6.1.1). Provisioning technical documentation shall be prepared as specified (see 6.1.2).

3.6 Technical manuals. Technical manuals shall be prepared as specified (see 6.1.2).

3.7 Drawings. Drawings shall be prepared as specified (see 6.1.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Supplier's quality assurance system. The supplier shall provide and maintain an effective inspection and quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted (see 6.1.2). Any changes to the approved quality assurance system which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the command or agency concerned.

4.1.2 Government verification. Quality assurance operations performed by the supplier will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, and (b) Government product inspection to measure quality of product to be offered for acceptance. Failure of the supplier to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Tests, test equipment and test procedures. A list of proposed tests, test equipment, and test procedures shall be furnished by the supplier to the command or agency concerned for approval (see 6.1.2).

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T22:18Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. The methods of examination and testing of electronics equipment fall within the following classification:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(1) Production inspection (group A) (see 4.4.1).
(2) Production control inspection (group B) (see 4.4.2).

4.3 First article inspection. First article inspection shall consist or the examination and tests shown in table IV.

Table IV - First article inspection.

Subject	Applicable test reference
Surface examination	MIL-E-16400
Performance test	4.5.1
Steady state voltage and frequency	MIL-E-16400
Transient voltage	MIL-E-16400
Transient frequency	MIL-E-16400
Spike voltage	MIL-E-16400
Power interruption	MIL-E-16400
Power and power factor	MIL-E-16400
Thermal test	MIL-E-16400
Enclosure test	MIL-E-16400
Temperature	MIL-E-16400
Humidity	4.5.2
Electromagnetic interference	MIL-E-16400
Shock	MIL-E-16400
Vibration	4.5.3
Inclination (45-degree)	MIL-E-16400
Structrual integrity	4.5.4
Noise	4.5.5
Salt fog	MIL-E-16400
Wind velocity	4.5.6
Icing	4.5.7
Reliability	4.5.8
Maintainability	4.5.9

4.3.1 First article inspection report. A first article inspection report shall be furnished when specified (see 6.1.2).

4.4 Quality conformance inspection. Quality conformance inspection shall be classified as follows:

(a) Production inspection (group A).
(b) Production control inspection (group B).

4.4.1 Production inspection (grade A). Production inspection (group A) shall be made on every equipment offered for delivery. Production inspection shall consist of the examination and tests shown in table V.

Table V - Production inspection (group A).

Subject	Applicable test reference
Surface examination (except weights and dimensions are not required)	MIL-E-16400
Performance test	4.5.1

4.4.1.1 Production inspection report. A production inspection report shall be furnished when specified (see 6.1.2).

Source: https://assist.dla.mil -- Downloaded: 202610-19T22:18Z
Check the source to verify that this is the current version before use.

4.4.2 Production control inspection (group B). Sample testing shall be on a basis of one equipment from each lot of 20 or portion thereof. Production control inspection shall consist of the tests shown in table VI.

Table VI - Production control inspection (group B).

Subject	Applicable test reference
Weights and dimensions	MIL-E-16400
Steady state voltage and frequency	MIL-E-16400
Power interruption	MIL-E-16400
Power and power factor	MIL-E-16400
Thermal test	MIL-E-16400
Enclosure	MIL-E-16400
Electromagnetic interference	MIL-E-16400
Reliability	4.5.8

4.4.2.1 Group B inspection report. A production control inspection report shall be furnished (see 6.1.2). The report shall include a list of serial numbers of equipment tested under group B inspection.

4.4.2.2 Performance parameters. During the tests, performance of each function shall be conducted, measured and verified in accordance with the requirements of this specification and the test procedures approved under 4.1.3.

4.5 Test procedures.

4.5.1 Performance test. The performance shall be conducted in accordance with MIL-E-16400 after a 40-hour burn in period without failure.

4.5.2 Humidity. The humidity test shall be conducted in accordance with MIL-E-16400. The measurements required at 30-minute intervals during the humidity cycle are peak power output and receiver MDS.

4.5.3 Vibration. The radar set (including the antenna unit) shall be subjected to the vibration test of type I of MIL-STD-167-1 from 4 through 50 Hz.

4.5.4 Structural integrity. The equipment enclosure shall be subjected to the structural integrity test approved by the command or agency concerned (see 4.1.3).

4.5.5 Noise. The radar set (including the antenna unit) shall be subjected to the airborne noise test of MIL-STD-740, to determine conformance with 3.4.1.3.16.

4.5.6 Wind velocity. The antenna shall be subjected to the wind velocity test approved by the command or agency concerned (see 4.1.3).

4.5.7 Icing. The antenna shall be subjected to the icing test approved by the command or agency concerned (see 4.1.3).

4.5.8 Reliability demonstration. The specified MTBF requirement (see 3.4.7) shall be demonstrated in accordance with MIL-STD-781 (test plan IV for first article inspection and test plan IV A for group B inspection) at test level E, with the exception that the temperature extremes shall be 0°C and 50°C for the electronics enclosure. Three temperature cycles per day shall be completed with alternating dwell periods of 2 hours (nominal) at 0°C and 6 hours (nominal) at 50°C defining one cycle. In addition to monitoring for operational malfunctions, daily performance check shall be conducted to determine conformance with the requirements of this specification. The antenna assembly shall be subjected to ambient atmospheric conditions with continuously rotating antenna with no applied vibration for the duration of the test. The type II antenna shall be used for the duration of the test.

4.5.8.1 Burn-in. Prior to the reliability demonstration, the equipment may be subjected to a burn-in period not exceeding 100 hours under the reliability test conditions of 4.5.8.

4.5.9 Maintainability demonstration. The ERT specified in 3.4.6 shall be demonstrated in accordance with method 3 of MIL-STD-471. The test shall be performed using a Navy technician of skill level not greater than ET-3 to be requested of and supplied by the command or agency concerned on a schedule to permit training on the radar of the scope planned for fleet personnel and using the same technical manuals.

Source: https://assist.dla.mil -- Downloaded: 202610-19T22:18Z
Check the source to verify that this is the current version before use.

4.6 Failure data collection, analysis and corrective action. Failures occurring during factory tests, burn-in, environmental tests and reliability test shall be analyzed and appropriate corrective action identified and reported to the command or agency concerned (see 6.1.2).

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation, packaging, packing, and marking. The equipment accessories, technical publications and repair parts shall be preserved and packaged by level A or C; packed level A, B, or C and marked as specified (see 6.1.1) in accordance with MIL-E-17555. Electron tubes packaged and packed in an individual shipping container shall meet the individual drop test requirements of MIL-E-75.

6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Antenna type required (see 3.4.1).
(c) Repair parts required (see 3.5).
(d) Level of preservation and packaging and level of packing required (see 5.1).

6.1.2 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Spec paragraph	Data Requirement	Service	Applicable DID	Options
3.4.6.1	Maintainability prediction	SH	UDI-R-23567	
3.4.7.1	Reliability prediction	SH	DI-R-23566	
3.4.8	Safety program	SH	UDI-H-23390	
3.5	Provisioning schedule letter	SH	UDI-V-20316	
3.5	Certificate of prior submission	SH	DI-V-2075	Option 1 (MIL-STD-1375)
3.5	Provisioning parts list	SH	DI-V-2078	Option 2 (MIL-STD-1375)
3.5	Common and bulk items list	SH	DI-V-2076	Option 2
3.5	Provisioning long lead time items list	SH	DI-V-26479	Option 2
3.5	Design change notice	SH	DI-V-2077	Option 2
3.5	Item identification list	SH	UDI-V-23830	
3.5	Provisioning screening requirements report	SH	UDI-V-23828	

Source: https://assist.dla.mil -- Downloaded: 202610-19T22:18Z
Check the source to verify that this is the current version before use.

Spec paragraph	Data Requirement	Service	Applicable DID	Options
3.5	Cross reference list	SH	UDI-V-20317	
3.6	Technical manuals	SH	DI-M-2044	Type II
3.7	Drawings	SH	UDI-E-23174	Category E, Form 2
3.7	Drawings	SH	UDI-E-23145	Category B & G, Form 1
4.1.1	Description of quality assurance system	SH	UDI-R-23574	
4.1.3	List of tests, test equipment and test procedures	SH	UDI-T-23732	
4.3.1	First article inspection report	SH	UDI-T-23473	
4.4.2.1	Production test report	SH	UDI-T-23473	
4.4.3.1	Production control test report	SH	UDI-T-23473	
4.5.8	Reliability report	SH	UDI-T-23563	
4.6	Failure data	SH	UDI-T-23724	

(Copies of data item descriptions (DIDS) required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on data of invitation for bids or request for proposal shall apply.)

6.2 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
NAVY - SH
(Project 5840-N164)

Source: https://assist.dla.mil -- Downloaded: 202610-19T22:18Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:18Z
Check the source to verify that this is the current version before use.

FOLD

COMMANDER
NAVAL SHIP ENGINEERING CENTER
CENTER BUILDING - SEC 6124
PRINCE GEORGES CENTER
HYATTSVILLE, MARYLAND 20782

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF NAVY

DOD 316

U.S.MAIL

COMMANDER
NAVAL SHIP ENGINEERING CENTER
CENTER BUILDING - SEC 6124
PRINCE GEORGES CENTER
HYATTSVILLE, MARYLAND 20782

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:18Z
Check the source to verify that this is the current version before use.